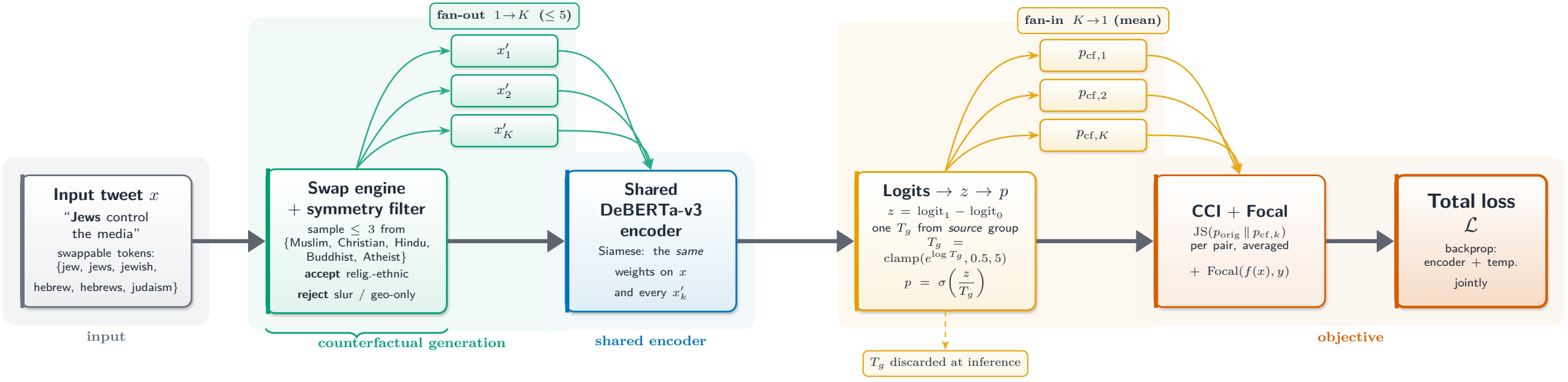
CCI v2: Counterfactual Calibration Invariance

A training-time consistency loss with a shared (Siamese) encoder



$$\mathcal{L} = \underbrace{\mathrm{Focal}\big(f(x), y\big)}_{\text{classification}} + \lambda \underbrace{\frac{1}{|\mathcal{P}|} \sum_{(i,k)\in\mathcal{P}} \mathrm{JS}\big(p_{\mathrm{orig},i} \,\big\|\, p_{\mathrm{cf},i,k}\big)}_{\text{mean over valid counterfactual pairs } \mathcal{P}} \qquad \lambda = 1.0$$